STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-FREE-GRADED-MODULE-GOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3128-3137 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3139-3183, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_sdga.json (pb=1574, stmt 3128-3137, proof 3139-3183).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSF: sources/stacks-sdga/pinned-original-source.tex, source lines 3098–3126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a differential graded $\mathcal{A}$-algebra. Let $\mathcal{S}$ be a sheaf of graded sets on $\mathcal{C}$. Then the free graded module $\mathcal{A}[\mathcal{S}]$ on $\mathcal{S}$ endowed with differential as in Remark 0FSF is a good differential graded $\mathcal{A}$-module.*

Proof. Let $\mathcal{N}$ be a left graded $\mathcal{A}$-module. Then we have

$$\mathcal{A}[\mathcal{S}] \otimes_{\mathcal{A}} \mathcal{N} = \mathcal{O}[\mathcal{S}] \otimes_{\mathcal{O}} \mathcal{N} = \mathcal{N}[\mathcal{S}]$$

where $\mathcal{N}[\mathcal{S}$ is the graded $\mathcal{O}$-module whose degree n part is the sheaf associated to the presheaf

$$U \longmapsto \bigoplus\nolimits_{s \in \mathcal{S}(U)} s \cdot \mathcal{N}^{n-\deg(s)}(U)$$

It is clear that $\mathcal{N} \to \mathcal{N}[\mathcal{S}]$ is an exact functor, hence $\mathcal{A}[\mathcal{S}$ is flat as a graded $\mathcal{A}$-module. Next, suppose that $\mathcal{N}$ is a differential graded left $\mathcal{A}$-module. Then we have

$$H^*(\mathcal{A}[\mathcal{S}] \otimes_{\mathcal{A}} \mathcal{N}) = H^*(\mathcal{O}[\mathcal{S}] \otimes_{\mathcal{O}} \mathcal{N})$$

as graded sheaves of $\mathcal{O}$-modules, which by the flatness (over $\mathcal{O}$) is equal to

$$H^*(\mathcal{N})[\mathcal{S}]$$

as a graded $\mathcal{O}$-module. Hence if $\mathcal{N}$ is acyclic, then $\mathcal{A}[\mathcal{S}] \otimes_{\mathcal{A}} \mathcal{N}$ is acyclic.

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Finally, consider a morphism $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ of ringed topoi, a differential graded $\mathcal{O}'$-algebra $\mathcal{A}'$, and a map $\varphi : f^{-1}\mathcal{A} \to \mathcal{A}'$ of differential graded $f^{-1}\mathcal{O}$-algebras. Then it is straightforward to see that

$$f^*\mathcal{A}[\mathcal{S}] = \mathcal{A}'[f^{-1}\mathcal{S}]$$

which finishes the proof that our module is good. □